Machine Learning: Midterm

Solutions

1. Evaluating classifiers (20 points)

A course runs an AI-text detector on all 400 essays submitted for an assignment. An honor-code review later established that 60 of the essays were written with AI. The detector flagged 50 essays, and 45 of the flagged essays were written with AI. Treat "written with AI" as the positive class.

1.1 [4 points] Write down the confusion matrix.

Solution: 60 AI-written essays and 340 human-written essays.

	flagged	not flagged
written with AI	TP = 45	FN = 15
human-written	FP = 5	TN = 335

1.2 [2 points] What are precision and recall for this detector?

Solution: Precision = $45/50 = 0.90$. Recall = $45/60 = 0.75$.

1.3 [2 points] Do we care more about improving precision or recall for this problem, and why?

Solution: Precision. A false positive accuses an innocent student of cheating; a missed AI essay (false negative) is a smaller harm. A flag should almost always be right before anyone acts on it.

The detector also gives each essay a score between 0 and 1; a higher score means "more likely written with AI". Here are 8 essays, 4 written with AI and 4 by students:

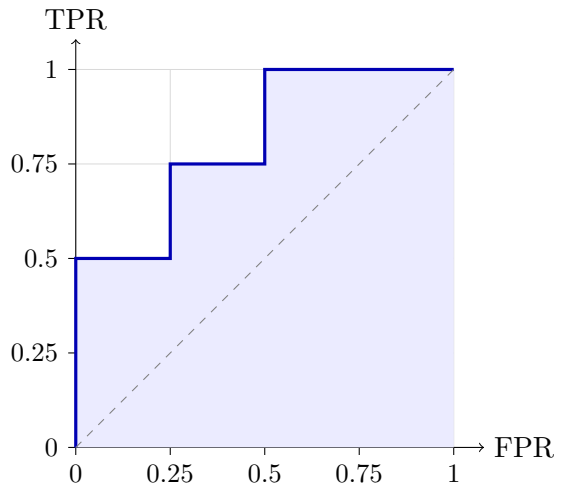
essay	1	2	3	4	5	6	7	8
score	0.95	0.85	0.70	0.60	0.50	0.40	0.30	0.10
written with AI?	yes	yes	no	yes	no	yes	no	no

We say the detector flags an essay as AI-written if its score is over t.

1.4 [10 points] Draw the ROC curve.

Solution: As t passes each essay, from the highest score down, an AI essay moves the point up by $1/4 = 0.25$ and a human essay moves it right by 0.25. Starting at $(0, 0)$, the points are

$$(0, 0.25),\ (0, 0.5),\ (0.25, 0.5),\ (0.25, 0.75),\ (0.5, 0.75),\ (0.5, 1),\ (0.75, 1),\ (1, 1).$$



1.5 [2 points] Compute the AUC.

Solution: The area under the staircase, in columns of width 0.25 with heights $0.5, 0.75, 1, 1$: $\text{AUC} = 0.25 \times (0.5 + 0.75 + 1 + 1) = 0.8125 \approx 0.81$.

2. Support vector machines (20 points)

2.1 [10 points] A soft-margin SVM returned $\boldsymbol{w} = (6, 8)$, $b = -20$.

point	A	B	C	D	E
x_1	3	1	1.5	1.25	3
x_2	1	1	1.5	1.5	0.5
label	+	−	+	−	−

(a) Compute $f(\boldsymbol{x})$ and the predicted class of each point.

(b) What is the width of the margin?

(c) Compute each point's distance to the decision boundary.

(d) Which points are support vectors? Which point is on the margin, which is inside it, and which is misclassified?

Solution: $f(\boldsymbol{x}) = 6x_1 + 8x_2 - 20$ and $\|\boldsymbol{w}\| = \sqrt{6^2 + 8^2} = 10$.

point	A	B	C	D	E
$f(\boldsymbol{x})$	6	−6	1	−0.5	2
predicted class	+	−	+	−	+
distance $\lvert f(\boldsymbol{x})\rvert/\|\boldsymbol{w}\|$	0.6	0.6	0.1	0.05	0.2
$y\, f(\boldsymbol{x})$	6	6	1	0.5	−2

(b) Margin width $= 2/\|\boldsymbol{w}\| = 2/10 = 0.2$.

(d) The support vectors are the points with $y\, f(\boldsymbol{x}) \le 1$: C, D and E. C is on the margin ($y\, f = 1$), D is inside the margin but correctly classified ($y\, f = 0.5$), and E is misclassified (label −, predicted +).

2.2 [4 points] The soft-margin SVM solves

$$\min_{\boldsymbol{w}, b, \boldsymbol{\xi}} \ \tfrac{1}{2}\|\boldsymbol{w}\|^2 + C \sum_i \xi_i \quad \text{subject to} \quad y_i(\boldsymbol{w} \cdot \boldsymbol{x}_i + b) \ge 1 - \xi_i, \ \ \xi_i \ge 0.$$

(a) What does the slack ξ_i measure, and why do we need it?

(b) Where does a training point lie, relative to the margin and the decision boundary, if $\xi_i = 0$? If $0 < \xi_i < 1$? If $\xi_i > 1$?

Solution: (a) ξ_i measures how far point i falls short of being on its correct side, outside the margin. Without slack, the constraints $y_i(\boldsymbol{w} \cdot \boldsymbol{x}_i + b) \geq 1$ have no solution when the classes overlap; slack lets some points violate the margin, at a cost $C\xi_i$ each.

(b) $\xi_i = 0$: on its correct side, on or outside the margin. $0 < \xi_i < 1$: inside the margin, but still correctly classified. $\xi_i > 1$: on the wrong side of the decision boundary (misclassified). (At the solution $\xi_i = \max(0, 1 - y_i f(\boldsymbol{x}_i))$, so these are the cases $y_i f \geq 1$, $0 < y_i f < 1$ and $y_i f < 0$.)

2.3 [4 points] Two soft-margin SVMs are trained on the same data, one with $C = 0.01$ and one with $C = 100$.

(a) Which has the wider margin, and why?

(b) Which has more support vectors, and why?

(c) Which is more likely to overfit, and why?

(d) How would you choose C in practice, and why?

Solution: (a) $C = 0.01$: slack is cheap, so the SVM accepts many points inside the margin in exchange for a wider one. (b) $C = 0.01$: with a wider margin, more points fall on or inside it, and all of them are support vectors. (c) $C = 100$: violations are expensive, so the margin narrows to fit individual training points. (d) C is a hyperparameter: try several values and pick the one with the best accuracy on validation data (or by cross-validation). Training accuracy cannot choose it, because a larger C always fits the training data at least as well; only held-out data shows which C generalizes.

2.4 [2 points] You remove a training point that is not a support vector and retrain the SVM. What changes? Why?

Solution: Nothing. The solution depends only on the support vectors; a point outside the margin does not affect $\boldsymbol{w}$ or b.